AI-POWERED INTELLIGENT EDUCATION SUPPORT SYSTEM

Group ID| 25-26J-402

SLIIT FACULTY OF COMPUTING

PP2 COMMENTS AND ACTION TAKEN

IT Number	Panel Comment	Action Taken
IT22907820	Should be prepared and come with the methodology to be explained during final presentation.	A well-prepared methodology enables clear, step-by-step explanation of the Attendance Trend Analyzer's predictive workflow during the final presentation
IT22590626	Was not able to show the model code itself. Should be prepared and come with the methodology to be explained during final presentation.	The development methodology was well-prepared and practiced to clearly explain the model workflow during the final presentation.
IT22565730	Was not able to show the model code itself. Should be prepared and come with the methodology to be explained during final presentation.	Instead of presenting the full source code, the system's development methodology was thoroughly prepared and practiced to clearly explain the model workflow and implementation process
IT22571984	Was not able to show the model code itself. Should be prepared and come with the methodology to be explained during final presentation.	Rather than relying on source code, the development methodology was well-prepared and practiced in advance to confidently explain the model workflow and implementation during the final presentation.

PROBLEMS IN BRIEF

Current Challenges in Education Monitoring

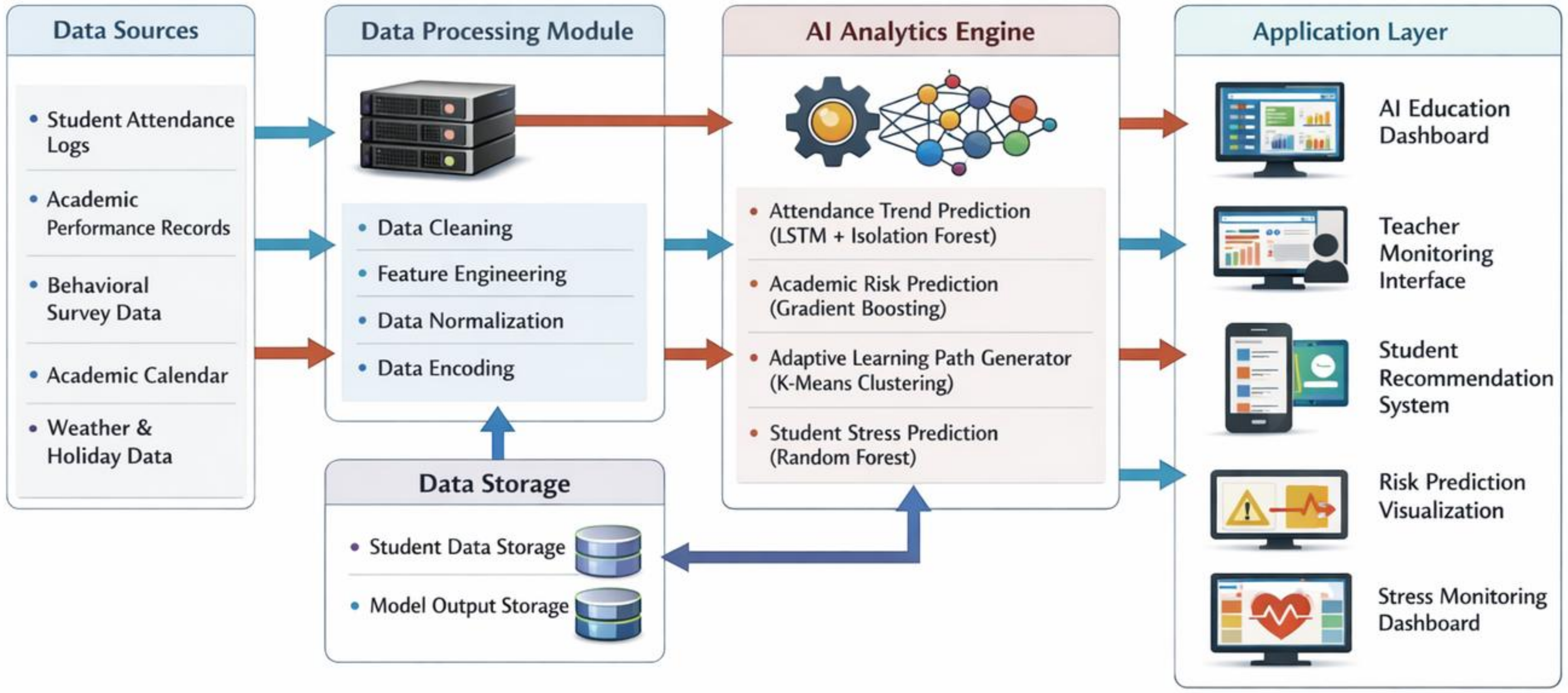
- Students face difficulty identifying academically struggling early.
- Student attendance patterns are often ignored until academic performance significantly drops.
- Students experience increasing academic pressure during O/L examinations.
- Lack of personalized learning recommendations for students with weak subject performance.
- Existing education monitoring systems rely heavily on manual observation rather than intelligent analytics.

SOLUTION OVEVIEW

- **Attendance Trend Analysis** – Analyzes historical attendance records to identify declining attendance patterns using time-series prediction models.
- **Academic Risk Prediction** – Uses machine learning models to classify students into Low, Medium, and High risk levels to identify struggling students early.
- **Adaptive Learning Path Generator** – Analyzes student academic performance and recommends personalized learning priorities for subjects that need improvement.
- **Student Stress Prediction** – Evaluates behavioral factors such as study hours, attendance trends, and academic performance to estimate student stress levels and provide support recommendations.

SYSTEM OVERVIEW DIAGRAM

Figure 1: EduGuide – AI Powered Intelligent Education Support System



ATTENDANCE TRENDS ANALYZER

KEY FEATURES

- AI-powered system that forecasts student attendance trends and detects unusual absence patterns using time-series analysis
- Applies advanced forecasting models (ARIMA & LSTM) to identify future attendance risks 2–4 weeks in advance
- Performs contextual correlation analysis with academic calendars, exam schedules, public holidays, and weather conditions
- Supports early identification of at-risk students before academic decline occurs
- Ensures privacy preservation through data anonymization and role-based access control

TECHNOLOGIES USED

- Time-Series Forecasting Models (ARIMA, LSTM) – Attendance trend prediction and forecasting
- Anomaly Detection (Isolation Forest) – Identify irregular attendance patterns and sudden drops
- Machine Learning Libraries (Scikit-learn, TensorFlow, Keras, Statsmodels) – Model training and implementation
- Backend Frameworks (Python Flask, Node.js/Express) – API services and backend processing
- Database (MongoDB) – Student attendance records, risk scores, and prediction history storage
- Frontend (React.js, Tailwind CSS) – Interactive and responsive web dashboard
- Data Visualization (Chart.js, Plotly) – Heatmaps, trend charts, and risk alert panels

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ACADEMIC RISK PREDICTION SYSTEM

KEY FEATURES

- AI system that predicts and classifies student academic risk levels using behavioral and performance data
- Uses student inputs such as attendance, average marks, study hours, homework completion, screen time, and study consistency
- Applies intelligent feature engineering for accurate prediction
- Provides risk level (Low / Medium / High), probability scores, and personalized improvement recommendations
- Supports early identification of at-risk students and helps improve O/L exam performance
- Uses machine learning models with explainable logic to ensure transparency and trust

TECHNOLOGIES USED

- Machine Learning Models (Random Forest) – Risk prediction & classification
- Feature Engineering – Trend analysis (performance, attendance, study behavior)
- Explainable Logic (Rule-based + Model Insights) – Understand why a student is at risk
- Flask API – Backend processing & prediction service
- MongoDB – Student data storage & prediction history
- React (Vite + Tailwind CSS) – User-friendly web interface
- Python (Pandas, NumPy, Scikit-learn) – Data processing & model training

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ADAPTIVE LEARNING PATH GENERATOR

KEY FEATURES

- Analyzes student academic performance to identify strengths and weak subject areas
- Uses clustering techniques to group students into learning categories (high, average, struggling learners)
- Generates personalized study plans and subject priorities for each student
- Recommends targeted learning resources and improvement strategies
- Helps students focus on weak areas and improve overall academic performance
- Provides continuous updates based on student progress

TECHNOLOGIES USED

- K-Means Clustering – Student segmentation
- Machine Learning Models – Learning recommendation generation
- Performance Data Analysis – Feature extraction from marks
- Recommendation Engine – Personalized study suggestions
- Flask API – Backend processing
- React Web Application – User interface (student dashboard)

STRESS LEVEL PREDICTION

KEY FEATURES

- Analyzes behavioral, academic and lifecycle factors to estimate student stress levels
- Uses inputs such as study hours, sleep duration, attendance, and performance trends
- Classifies students into stress levels (Low / Moderate / High)
- Identifies key factors contributing to stress using feature importance
- Provides personalized recommendations to reduce academic stress
- Supports early intervention to improve student well-being

TECHNOLOGIES USED

- Random Forest Classifier – Stress level prediction
- Machine Learning Models – Behavioral data analysis
- Feature Importance Analysis – Identifying key stress factors
- Data Preprocessing – Cleaning and normalization
- Flask API – Backend integration
- React Web Application – Stress monitoring dashboard

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RESULT AND TESTIMONIALS

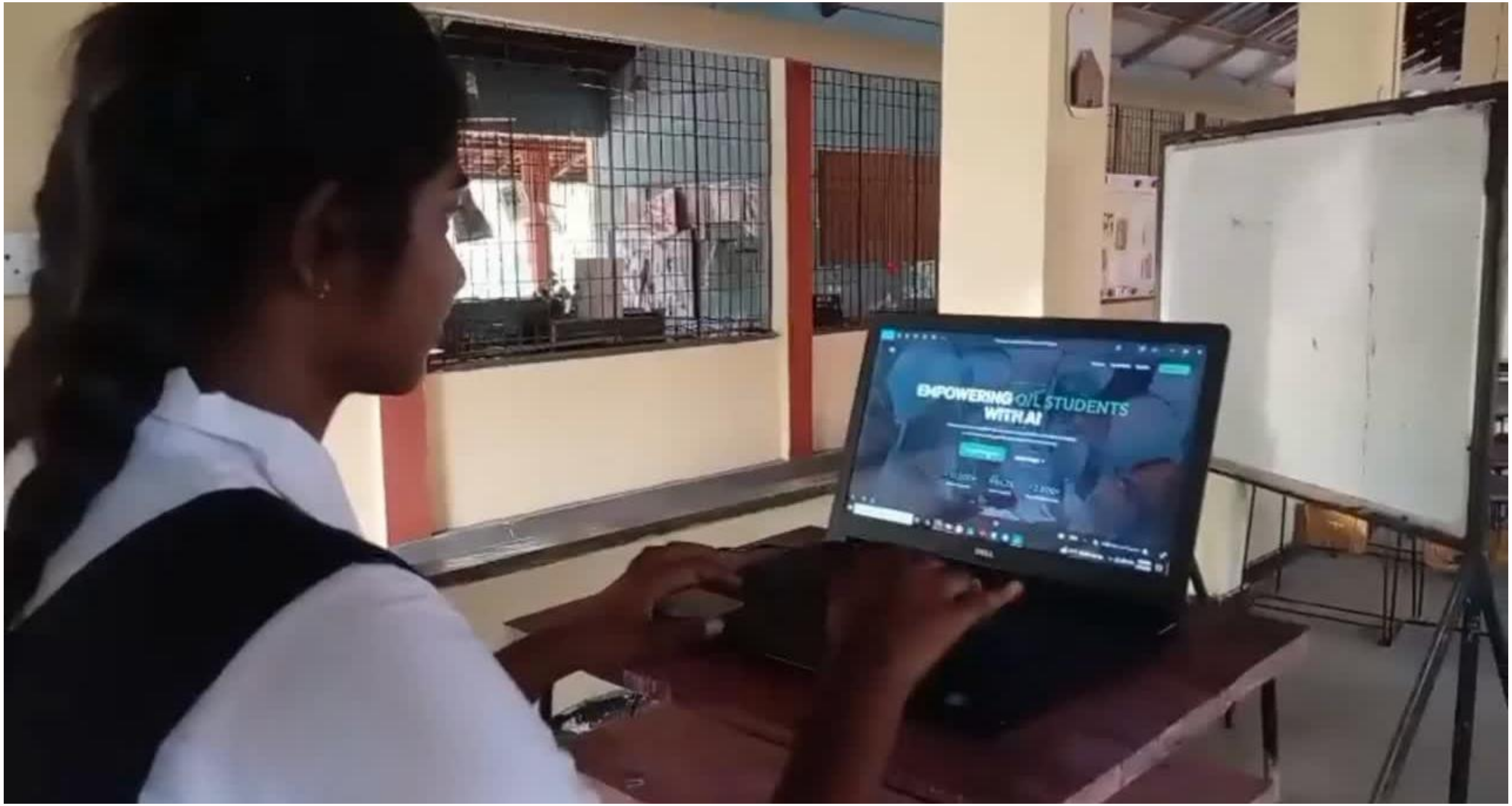
RESULTS

- Academic Risk Prediction model achieved high accuracy in identifying at-risk students
- Attendance analysis successfully detected declining attendance patterns early
- Adaptive learning module provided personalized subject recommendations based on student performance
- Stress prediction component effectively classified student stress levels using behavioral data
- System provided real-time insights through an interactive web dashboard

KEY ACHIVEMENTS

- Early identification of low-performing students
- Improved awareness of student attendance and behavior patterns
- Personalized learning support for students
- Data-driven decision-making support for users

FEEDBACK



LIMITATIONS

- Model performance depends on the quality and accuracy of student data (attendance, marks, surveys)
- Predictions may be affected by incomplete or missing student information
- Limited dataset size may reduce model generalization and accuracy
- System performance depends on stable internet connectivity (web-based application)
- Behavioral data such as stress indicators rely on user-provided inputs, which may not always be accurate
- External factors (family, environment, personal issues) are not fully captured by the system
- Predictions are probabilistic and not guaranteed to be 100% accurate

FUTURE IMPROVEMENTS

- Introduce offline or low-connectivity support for rural schools
- Enhance prediction reliability using advanced ensemble and deep learning techniques
- Integrate additional data sources (exam systems)
- Develop a mobile application for better accessibility
- Add multilingual support (Sinhala / Tamil / English) and voice-based interaction
- Incorporate real-time data tracking and continuous model learning

THANK YOU